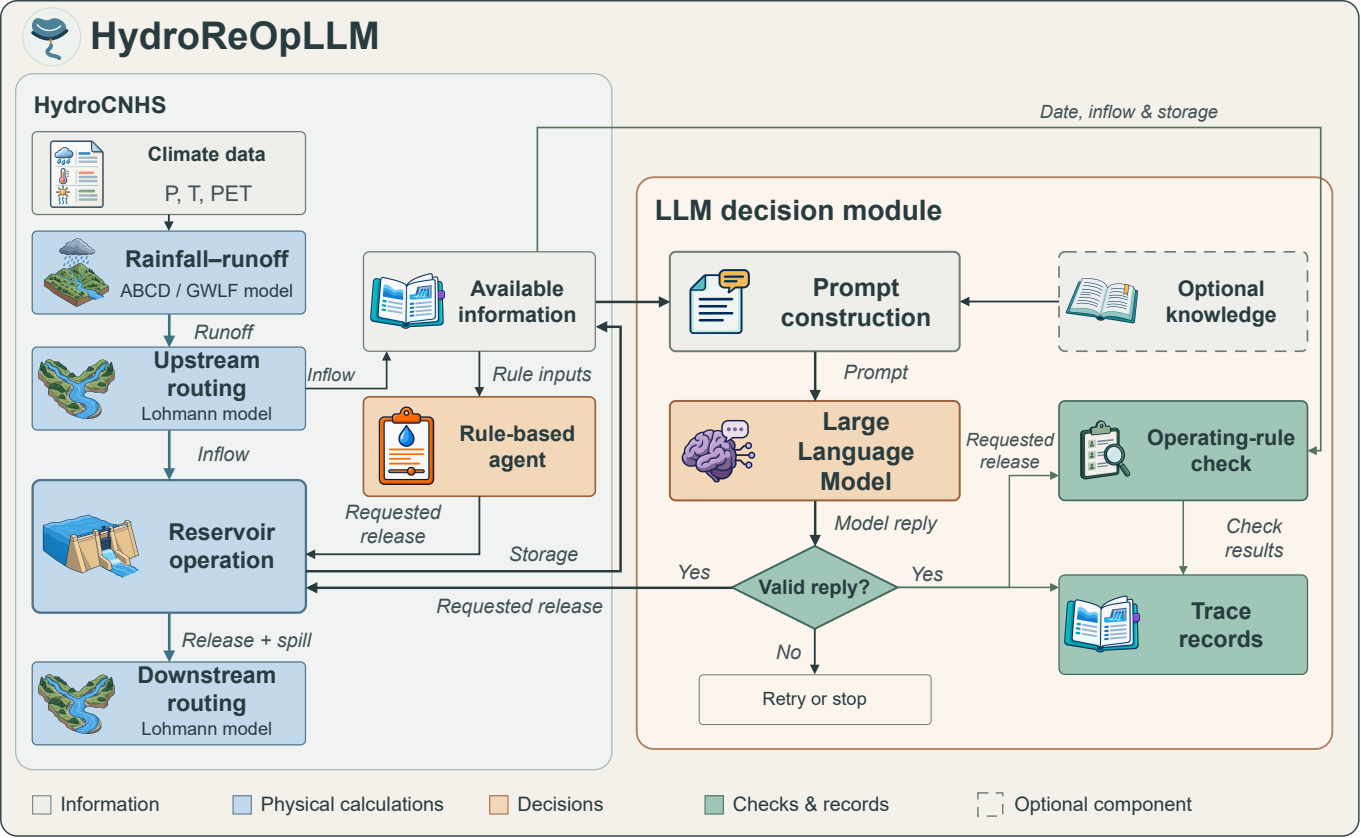

HydroReOpLLM
HydroCNHS
Climate data
P, T, PET
Rainfall–runoff
ABCD / GWLF model
Runoff
Upstream routing
Lohmann model
Inflow
Inflow
Reservoir operation
Release + spill
Downstream routing
Lohmann model
Available information
Rule inputs
Rule-based agent
Requested release
Storage
Requested release
Date, inflow & storage
LLM decision module
Prompt construction
Optional knowledge
Prompt
Large Language Model
Model reply
Valid reply?
Yes
Yes
No
Retry or stop
Requested release
Operating-rule check
Check results
Trace records
Information
Physical calculations
Decisions
Checks & records
Optional component